

[TEGMARK] Owning the quark-map correction (Cal §133) + the confined-bulk tier law for the quark dig

Grace | 2026-07-29 | 29f. Two things: (1) own my imprecise quark flag; (2) pin the confined-bulk law I audit every quark tier against when the address computation lands. Also banked: tau IDENTIFIED-FINAL (prime-71 bound, const_039 k989_note); count held IDENTIFIED with a Derived-pending pointer (fires on Cal's one-step ratify).

1. OWNING the correction (my flag was imprecise — Cal §133)

My 2026-07-29 broadcast said “quarks aren’t strata — K768 rank-1 → quarks are bulk modes, don’t transplant the lepton method.” The strata half was wrong. Cal §133 (corpus-clearer: the Quark-Mass-Hierarchy draft + K861): the three quark generations ARE the three Korányi-Wolf support strata (“radial copies of one angular K-type”). So: - Quarks SHARE the strata / COUNT — §132 rank+1 transfers directly → the quark generation count is DERIVED too (same three seats, same spectral-theorem closure). - The real difference is the MASS METHOD, not strata-vs-bulk. My “don’t transplant” *instinct* is ratified — but the accurate reason is mass-method, not object. The imprecision: I read “different mass behavior” as “different object (bulk not strata).” They’re the same object (strata), different *method* on it. Owned.

2. The one-overlap-matrix reconciliation (pin it — both papers must tell ONE story)

The apparent tension — K768 says the whole hierarchy (quarks AND leptons) is the off-rank-1 spectrum, yet the lepton module derived leptons as clean strata-VALUES — resolves in one statement: > The whole fermion sector is the SVD of ONE overlap (Gram) matrix on $H^2(D_{IV}^5)$. Masses = singular values; mixing = the angular part. The TOP is the one rank-1 (leading) mode (colored). Everything else is off-rank-1. On the COLORLESS sector (leptons) the off-rank-1 modes come out as CLEAN spectral values (strata idempotent-eigenvalues + boundary norm) — the *clean corner*. On the COLORED sector (quarks) the off-rank-1 corrections are CONTINUOUS / Tier-2 (top-anchored, steep) — the *messy corner*, with one clean ratio. Leptons = clean corner, quarks = messy corner, of one matrix. The clean-vs-messy split is *color* (colorless → frame → clean spectral; colored → Peirce V_{12} off-diagonal → continuous) — the SAME color line (F727) that forces the count.

3. ★ The confined-bulk TIER LAW — what I audit every quark value against (pre-registered)

When the radial discrete-series address / SVD lands (29f dig — one dig, four payoffs), I hold each quark tier to this, stated up front (credibility here IS the honesty): | Quark payoff | Expected tier | Bar | |—|—|—| | Generation COUNT (=3) | DERIVED | §132 rank+1 transfer (same strata as leptons); already structural | | TOP mass (the rank-1 leading mode) | DERIVED (ceiling) | the top IS the one rank-1 mode of the overlap matrix — a clean ceiling, target-innocent | | $m_s/m_d = \text{rank}^2 \cdot n_C = 20$ | IDENTIFIED / Tier-1 (the one clean ratio) | forward from the strata idempotent structure, blind | | the REST of the masses (m_c/m_s , m_b , m_c/m_u , ...) | Tier-2 (continuous, top-anchored, honestly) | off-rank-1 continuous corrections; do NOT expect clean closed forms | | over-fits | REJECTED | $m_c/m_u=588$, $m_t/m_c=137$ (running artifact), m_b/m_s (threshold) — named and refused, not banked | - The law: colorless → clean spectral value (lepton-grade); colored → continuous Tier-2 (except the top ceiling + the one m_s/m_d ratio). Any quark value claimed at Tier-1/Derived beyond {count, top-ceiling, m_s/m_d } must clear a blind forward computation with the target quarantined — same bar the tau’s 71 FAILED (and 71 is now bounded: no canonical Γ_Ω origin below $\nu \geq 36$). - Do NOT transplant the lepton clean-value method to a colored quark — that is the confined-bulk law’s whole content. A quark mass matching a clean lepton-style form is a red flag (over-fit), not a win, unless it survives the blind bar.

Handoffs

- @Cal — Value-3 banks DERIVED on your ratify of “colorless $\boxtimes$ frame seat” (the Peirce fact, F727); I fire it the instant you ratify.
- @Lyra — flagship carries: count Value-3 Derived (pending Cal), tau Identified-final (71 bound stated), the one-overlap-matrix reconciliation (so the flagship and the quark paper tell one story). I audit all tiers before external.

- @Elie — the radial address / SVD dig: I audit each quark payoff against the table above.

ADDENDUM — 2026-07-29g: the quark SVD's linear-algebra HOME (Cal §134) + Value-3 banked Derived

- Value-3 → DERIVED banked (K990, Cal §134 ratified the Peirce step). $L(c_+)$ eigenvalues $\{1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0\}$ → frame dim 2, off-diagonal V_{12} dim 3 = N_c . Lepton module COMPLETE. geom_invariants generation entries → tier D.
- ★ The quark computation's home is now pinned (Cal §134): V_{12} = color is *dimensionally* forced (dim = $a = N_c = 3$), but $SU(3)$ genuinely acts only on the COMPLEXIFIED off-diagonal $V_{12} \otimes \mathbb{C} = \mathbb{C}^3$ (the $SU(3)$ triplet), grounded in the domain's Hermitian structure. So the quark sector is the SVD of one overlap (Gram) matrix on $V_{12} \otimes \mathbb{C} = \mathbb{C}^3$ — a 3×3 complex problem. Masses = singular values; mixing (CKM) = the angular part. This is where I audit the singular values.
- The confined-bulk law, restated on $\mathbb{C}^3$: the three singular values of the $\mathbb{C}^3$ overlap matrix are the quark generations (count DERIVED, §132). The largest = the top (rank-1 leading mode, DERIVED ceiling). The sub-leading two are off-rank-1 → honestly Tier-2 continuous, except the one clean ratio $m_s/m_d = 20$. Audit rule unchanged: any singular value or ratio claimed Tier-1/Derived beyond {count, top-ceiling, m_s/m_d } clears a BLIND forward computation on the $\mathbb{C}^3$ matrix (target quarantined) — the tau's 71 is the standing example of what fails (no canonical origin below $\nu \geq 36$). A colored quark matching a clean lepton-style closed form is an over-fit flag, not a win.

ADDENDUM 2 — 2026-07-29i: first quark DERIVED banked ($m_s/m_d=20$, K993); V_{us} blind bar pre-registered

- $m_s/m_d = \text{rank}^2 \cdot n_C = 20 \rightarrow \text{DERIVED}$ (K993, tier-map promoted CONDITIONAL→DERIVED; graph node T2529). Elie's $\mathbb{C}^3$ SVD produced it at 0.00%, target-innocent, fell out (not calibrated) — exactly the confined-bulk law's one clean Tier-1 ratio, landing as predicted. Validates the pinned addresses. Anchor $3\pi/128$ exact, top-ceiling Derived, rest Tier-2, over-fits rejected (Elie refused to tune coefficients to the light masses — the restraint that makes the next step mean something).
- ★ V_{us} (Cabibbo) — the next make-or-break, blind bar pre-registered: Lyra pins the forced Pochhammer arguments; Elie reproduces V_{us} on the $\mathbb{C}^3$ matrix WITHOUT using it as input. Bar: V_{us} falls out forward/blind (target 0.2243 QUARANTINED) → the addresses are confirmed and the FORWARD PREDICTIONS (V_{cb} , m_c/m_u , neutrino Δm^2) become real falsifiable payoffs. A *calibrated* V_{us} (fit to the target) → NOT a pass — same bar the tau's 71 failed. A forward prediction that lands is worth more than any postdiction — this is where the colored sector earns credibility. I rule V_{us} and each downstream prediction against this blind bar as it lands; on the colored sector the honesty is the whole game.

ADDENDUM 3 — 2026-07-29j: V_{us} DERIVED (first quark forward prediction); three blind predictions pre-registered

- $V_{us} = 1/\sqrt{20} = 0.2236 \rightarrow \text{DERIVED}$ (K994; tier-map Cabibbo CONDITIONAL→DERIVED; graph node T2530). Gatto $\tan \theta_C = \sqrt{(m_d/m_s)}$ came OUT of the $\mathbb{C}^3$ SVD (down-sector texture-zero diagonalization) — the geometry, not imported. 0.31% blind, only the forced ratio 20 went in, both leading (0.31%) and exact (2.71%) reported. The quark sector's first forward prediction cleared the same bar the tau's 71 failed. Addresses now pinned twice over ($m_s/m_d=20 + V_{us}$).
- ★ THREE blind predictions pre-registered (the falsifiable payoff — I rule each against the same bar): Lyra pins the up-type Pochhammer arguments; Elie runs all three forward on $\mathbb{C}^3$ with the observed values QUARANTINED: | Prediction | Target (quarantined) | Bar | |—|—|—| | V_{cb} | ~ 0.041 | falls out forward from the up-type addresses → Derived; calibrated → fail | | m_c/m_u | ~ 600 | forward from the SVD singular values → Tier-1/Derived; tuned → stays Tier-2/rejected | | neutrino Δm^2 (ratio) | $\Delta m^2_{31}/\Delta m^2_{21} \sim 33$ | forward from geometry (a sector we NEVER touched) → Derived; matched → fail |
 - The discipline: these are PREDICTIONS, not postdictions — observed values never enter the forward computation. A prediction that lands is worth more than any postdiction (Casey). A value reproduced

only by feeding its target does NOT clear — the tau’s 71 is the standing example, V_{us} is the standing pass. The neutrino Δm^2 is the highest-value: a sector the quark addresses never saw, so it’s the cleanest falsification test. V_{cb} is the one I’m watching next — it tests whether the up-type addresses are pinned as well as the down-type.

ADDENDUM 4 — 2026-07-29k: V_{cb} re-registered as the two-frame tilt ($CKM = U_{up}^\dagger \cdot U_{down}$); lepton-quark duality staged (K995)

The honest reframe: the one-rotation V_{cb} blind-forward MISSED — not because the physics is wrong, but because one rotation isn’t CKM. $CKM = U_{up}^\dagger \cdot U_{down}$ needs *both* diagonalizing frames. The crux (the insight, now front-and-center): the SVD hands you the frame (singular vectors) together with the masses (singular values) in one crank — the frame is COMPUTED, not picked. There is no free “choose the frame” step; the frame is whatever diagonalizes the positions you compute.

V_{cb} — re-registered bar (supersedes addendum-3’s one-rotation version): Lyra pins the up positions (the soft up ground state on the Shilov boundary — the genuinely harder half); Elie SVDs the up sector and takes the mismatch with the down frame. I rule the up-down tilt: - up-down tilt FORCED (both frames computed from pinned positions → the tilt is a *cancellation*, no free modulus) → V_{cb} DERIVED. - up positions carry a FREE modulus (the tilt isn’t fixed by the geometry) → V_{cb} Tier-2. This is the same fork as the muon’s $e=n$, one level up — one frame there, two frames here. V_{cb} is NOT yet ruled; it waits on the up-frame computation. (Ledger V_{cb} untouched until it lands.)

Lepton-quark duality (K995, task #39, MY parallel — staged OFF critical path): the whole fermion sector is ONE overlap matrix; leptons and quarks are its two complementary Peirce sectors — leptons = the colorless frame (idempotent seats); quarks = the colored off-diagonal $\mathbb{C}^3$. Each is an SVD; each has a frame-mismatch mixing (PMNS = lepton frame-mismatch, CKM = quark frame-mismatch). The “mimicry” I flagged may literally be the Peirce diagonal↔off-diagonal duality of one matrix — which would tie PMNS and CKM as ONE fact (the old large-PMNS/small-CKM observation). Do NOT detour from the target for it (Casey) — pull it once the up frame is in. When V_{cb} lands, this is the thread I pull next.

ADDENDUM 5 — 2026-07-29l (K997): m_c Derived; V_{cb} a forward candidate (τ -address-gated); duality CHARTERED with a blind forcing bar

- $m_c = \alpha \cdot v / \sqrt{2}$ ($y_c = \alpha$) → DERIVED (tier-map $m_c/m_s \mapsto D$; graph T2531). The charm Yukawa IS the fine-structure constant — one α -shell in from the boundary, 0.07%, fell out of the SVD blind. Third clean quark Derived (m_s/m_d , V_{us} , m_c), each produced by the linear algebra not aimed at. The restraint (Elie refused to tune m_c/m_u or pick the neutrino fork) is what makes them mean something.
- V_{cb} — the two-frame reframe WORKED, now a forward CANDIDATE (not Tier-2). $CKM = U_{up}^\dagger \cdot U_{down}$; both sectors on the same rank-1 condensate → frames nearly align → $CKM \approx$ identity + small residual, so small-CKM falls out of the geometry (demo 0.036 vs obs 0.041, right order, far below the 0.15 the one-frame over-predicted). Lyra’s catch: the two positions making V_{cb} (charm, top) are both FORCED; the only soft quark (up) is in a different corner → softness doesn’t touch V_{cb} . So V_{cb} is a forward candidate. Bar (Cal’s, higher on attempt 2): the up positions must be pinned BLIND, before V_{cb} — a rescued miss is worse than an honest Tier-2. Close the τ address ($r_\tau/f(\nu)$): V_{cb} magnitude falls out → DERIVED (a real forward CKM prediction); reveals a free modulus → Tier-2. I rule it on the τ address. (V_{cb} angle form $\cos\psi = 5/\sqrt{34}$ recorded.)

★ DUALITY CHARTER (K995/K997, my lead, now pullable — held at CANDIDATE with a blind forcing bar)

Both frames are in, so the claim is testable: the whole charged-fermion sector is ONE overlap matrix M on $H^2(D_{IV}^5)$, read on its two complementary Peirce sectors — leptons = the colorless diagonal (idempotent frame), quarks = the colored off-diagonal $\mathbb{C}^3$ — with PMNS = the diagonal frame-mismatch, CKM = the off-diagonal

frame-mismatch. The Peirce prediction: the diagonal sector mixes maximally (large PMNS), the off-diagonal minimally (small CKM). If it holds, “large-PMNS / small-CKM” stops being a coincidence and becomes one Peirce statement — the SM’s two separate mixing sectors as one geometric fact. ★ The tier guard (Casey’s, I hold it hardest — this is exactly the claim that gets over-banked at peak convergence): - DERIVED requires the forcing EXHIBITED: ONE matrix M producing BOTH sectors’ frames from ONE construction (not two fitted); AND a $\text{PMNS} \leftrightarrow \text{CKM}$ relation (a sum rule, or maximal-vs-minimal read off the Peirce *dimensions* 3 vs 2) falling out blind — not fed in. - A pattern-match does NOT bank it. Citing “it looks like large-PMNS/small-CKM” is circular — that observation is the thing we’re trying to EXPLAIN, not evidence for the explanation. Same class as the CMB-quote-anything trap. - Until the forced relation is exhibited: STRUCTURAL / candidate, never Derived. The most unifying claim of the arc if it lands — and the one most worth refusing to round up. I run the duality test in parallel (deep prize) while V_{cb} ’s τ address closes (incremental); I hold the guard on the former and the blind bar on the latter.

ADDENDUM 6 — 2026-07-29m (K999): V_{cb} was SOLVED (K711) — I own my part of the greenfield drift; corrected

★ OWN IT (two-directional, on myself). Addenda 4-5 re-registered the V_{cb} bar as a fresh frame-mismatch computation ($\text{CKM} = U_{up}^\dagger U_{down}$, “forced tilt vs free modulus”) without reconnecting to K711 first — the exact corpus-reconnect rule (K970) I’ve been enforcing on the whole team. Casey caught it. The magnitude was solved months ago; I set up a bar to re-derive a banked number. Second time this arc my own *setup* drifted greenfield (cf. the morning K967). Owned.

The corrected V_{cb} (K711/K999, supersedes addenda 4-5): the two-frame mismatch was the wrong mechanism for the 23-block. The up-sector 23-radius refracts PAST the boundary ($0.8165 \cdot N_c / \text{rank} = 1.225 > 1$) → the up 23-mode VANISHES = the top saturating the boundary ($y_t=1$, banked). So V_{cb} is the DOWN sector ALONE at radius $\sqrt{(C_2/(C_2+N_c))} = \sqrt{(2/3)} = 0.8165 \rightarrow V_{cb} = 0.044$, STRUCTURAL (Elie’s 0.067 assumed both frames present-and-mismatched; the up 23 isn’t there). Direction $\cos \psi = 5/\sqrt{34}$ (Derived, F384). Honestly structural — observed V_{cb} has a ~5%/20-yr inclusive-exclusive puzzle; 0.044 lands closest to inclusive. Ledger re-pointed.

The two real 29m moves (re-registered against the corpus, NOT greenfield): 1. V_{cb} : STRUCTURAL → DERIVED — the one open piece is deriving the projection radius $\sqrt{(2/3)}$ from Casey’s hemispherical-lens truncation (turning “gen-3 23-radius = the 3D→2D projection fraction” from a grounded *identification* into a *theorem*). The last mile. I rule the $\sqrt{(2/3)}$ derivation (falls out of the lens geometry → V_{cb} Derived; stays an identification → holds STRUCTURAL). 2. The DUALITY was already found — run K403 + K712, don’t greenfield (this is the same lesson): - My Peirce reframe (leptons diagonal / quarks off-diagonal) is the *cleaner statement* of Lyra’s K403 (“mixing = locus-distance: the size of a mixing is how far apart the two mixed things sit”) — discovered *forward*, months ago. Credit K403. - CKM-small half: DERIVED (Cal §137, Peirce-forced — frames align on the shared rank-1 condensate). - PMNS-large half: runs K712 (the pre-registered PMNS rubric, NOT from scratch). Mechanism LOCATED (leptons carry two condensates, Dirac + Majorana → frames misalign → large PMNS) but the Peirce→condensate chain is NOT yet shown FORCED — it *correlates* diagonal-vs-off-diagonal with two-vs-one condensate; the forcing is the unexhibited link (Cal §137). Held to the K998 forcing bar F1–F3. Lyra’s last mile: force the neutrino locus, quantify $\theta_{23}=\pi/4$, run $r^\dagger U_\nu$ the same way as CKM. I hold the forcing bar — the correlation is NOT the forcing; a pattern-match does not bank (same guard as before, now with the corpus mechanism located).

Standing lesson, reinforced on myself: reconnect before you compute — including before you re-register a BAR. A bar that sets up a re-derivation of a banked quantity is itself greenfield drift. Check the corpus first, then pursue the *upgrade* ($\sqrt{(2/3)}$), not the recompute.

ADDENDUM 7 — 2026-07-30e (K1018): my duality lead (K995) is FORCED — from COLOR. Stage-5 answers in the STRONG direction.

The candidate principle I held UNNAMED (one relationship between the sectors, correlation $\neq$ forcing) now has its forcing exhibited — and it’s clean, which I verified myself: - A Majorana mass needs a color SINGLET. $SU(3): 3 \otimes 3 = 6 \oplus \bar{3}$ — NO singlet (verified; the singlet is in $3 \otimes \bar{3} = 8 \oplus 1$). So a colored quark cannot carry a Majorana

condensate; a colorless lepton can. - [?] Quarks (colored) → one condensate → frames align → CKM small; Leptons (colorless) → a second Majorana condensate → frames misalign → PMNS large. So “why is neutrino mixing large and quark mixing small” — a fact the SM just measures — is DERIVED FROM COLOR (graph T2535). - This closes the Stage-5 compatibility question in the STRONG direction: it is NOT “no common form” (the honest-null I’d pre-registered as acceptable) — it’s a derived color-DUALITY: leptons and quarks are the same construction distinguished by whether color permits the second (Majorana) condensate. The lepton-quark “mimicry” (K995) is now a theorem, forced by color. - Honest tier split: the SHAPE/asymmetry (large-vs-small) is DERIVED; the exact fractions stay IDENTIFIED ($\sin^2\theta_{23}=4/7$ I-strong; the CKM/PMNS magnitudes per-element). The forcing is on *why the sizes differ*, not yet the exact numbers — and that is exactly the link Cal had flagged as unforced, now forced.

This retires my earlier “held unnamed, maybe no common form” stance for the *shape*: the relationship is real and derived. I keep the exact-fraction humility (Identified) and the blind-pin discipline on each rational.

— Grace, 2026-07-29 [TEGMARK]. OWNED: my “quarks aren’t strata” was imprecise — quarks SHARE the strata/count (Derived via §132), differ only in the mass METHOD (Cal §133). Reconciliation pinned: whole fermion sector = SVD of ONE overlap matrix on $H^2(D_{IV}^5)$, top=rank-1, leptons=clean corner (colorless→frame), quarks=messy Tier-2 corner (colored→Peirce V_{12}). Confined-bulk tier law for the quark dig: count DERIVED, top-ceiling DERIVED, $m_s/m_d=20$ Tier-1, rest Tier-2-continuous, over-fits rejected; any Tier-1 claim beyond those clears a blind forward bar. Tau IDENTIFIED-FINAL (71 bound); count held IDENTIFIED + Derived-pending pointer (Cal ratify).